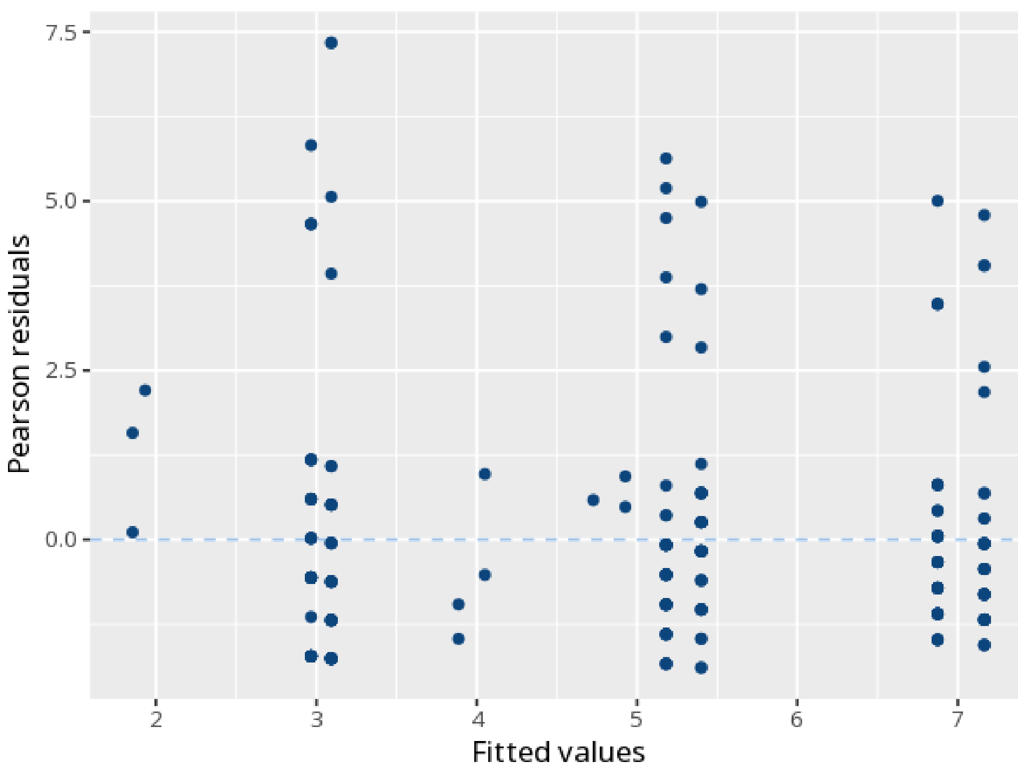
1 Poisson / negative binomial

	Log-count (B)	IRR
(Intercept)	−1.09	0.34
	(0.06)	[0.30, 0.38]
teaching_method: Flipped	−0.56	0.57
	(0.08)	[0.49, 0.67]
teaching_method: Lecture	0.28	1.33
	(0.06)	[1.17, 1.51]
gender: Male	−0.04	0.96
	(0.06)	[0.86, 1.07]
Num.Obs.	240	
Dispersion	3.14	
Residual deviance	633.11	
df	236	
Log.Lik.	−683.51	
AIC	1375.03	
BIC	1388.95	



Models event counts. Each predictor's incidence-rate ratio (IRR) gives the multiplicative change in the expected count per unit (>1 more, <1 fewer); p tests significance. Check dispersion to pick Poisson vs. negative binomial. If cases have unequal exposure (different observation time/area/population), add an offset of log(exposure) so the model predicts rates, not raw counts — omitting it biases the IRRs. Your significance threshold (α) is 0.05.

Predictor teaching_method: Flipped was associated with the count, IRR=0.57, 95% CI [0.49, 0.67], $p < .001$.

Statistical basis: Poisson regression (incidence-rate ratios) (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Negative binomial regression (glm.nb) for overdispersed counts (Venables WN, Ripley BD (2002). Modern Applied Statistics with S, Fourth edition. Springer, New York.); Overdispersion check (Lüdecke D, Ben-Shachar MS, Patil I, Waggoner P, Makowski D (2021). "performance: An R Package for Assessment, Comparison and Testing of Statistical Models." Journal of Open Source Software, 6(60), 3139.).